Angular momentum deficit and exponentially long collision safety in the planetary problem

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November 1, 2026

Abstract

Laskar and Petit conjectured that a planetary system whose angular momentum deficit (AMD) is too small to allow orbits to touch is, away from short-period resonances, stable in practice. We prove a rigorous asymptotic form of the mechanism behind this conjecture, in the plane and in space, up to the strong threshold, which also forbids a planet from reaching the star and is geometrically sharp. Fix the normalized masses, a compact set of ordered semi-major axes and a margin $\delta > 0$ below the threshold. If the planetary masses are of order μ, with μ small, and the initial mean motions are Diophantine with exponent τ, then for $|t| \le T_0 \exp(c\mu^{-1/(2(\tau+1))})$ the semi-major axes move by $O(\sqrt{\mu})$, the AMD stays at least $\delta/2$ below the threshold, and the orbital annuli stay ordered and uniformly separated. Without arithmetic conditions the same holds with Nekhoroshev's exponents: drift $O(\mu^{\mathfrak{b}_n})$ for $|t| \le T_0 \exp(c\mu^{-\mathfrak{a}_n})$. Exact conservation of angular momentum confines the AMD while the semi-major axes drift slowly; the secular dynamics may be chaotic. The constants are not explicit. All results are formally verified in the Lean proof assistant.

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1 Introduction

Consider a star and $n \geq 2$ planets whose masses are a small multiple μ of the mass of the star, interacting by Newtonian gravity. To first approximation each planet follows a Kepler ellipse. The mutual attractions deform these ellipses: over times of order μ^{-1} the eccentricities, inclinations and orientations change appreciably (the secular evolution), while the semi-major axes stay nearly constant. Whether the orbits can come to intersect, so that close encounters, collisions or ejections become possible, depends on how far the eccentricities can grow. In this paper we prove that an elementary conserved quantity bounds this growth for an exponentially long time, with a threshold that is geometrically sharp: at the threshold, two orbits, or an orbit and the star, can touch.

The angular momentum deficit. The *angular momentum deficit* (AMD) of a planetary system is the difference between the angular momentum that the planets would have on circular, coplanar orbits with the same semi-major axes and their actual angular momentum [7, §2.2]. In units where the gravitational constant and the mass of the star are one, and with planet masses $\mu\nu_i$, the Delaunay actions are $\mu\Lambda_i$ with $\Lambda_i = \nu_i\sqrt{a_i}$, and the AMD reads $\mu\sum_i \Lambda_i(1-\sqrt{1-e_i^2}\cos\iota_i)$ [7, (11)], where a_i, e_i and ι_i are the semi-major axes, eccentricities and inclinations relative to the plane orthogonal to the total angular momentum. The AMD vanishes exactly for circular coplanar prograde orbits, and a planet can reach large eccentricity only if the AMD is large enough. Laskar and Petit [7, §3] computed the least AMD at which the orbits of two planets with given semi-major axes and masses can touch. They call a system *AMD-stable* if its AMD is below this value for every adjacent pair, and *strongly* AMD-stable if, moreover, it is below the circular angular momentum $\mu\Lambda_1$ of the innermost planet, the value at which this planet can reach the star [7, §5 and §5.3]. Since the AMD is conserved by the secular dynamics at all orders, they formulate the following conjecture [7, §3]:

> *we conjecture that in absence of short period resonances, the AMD-stability ensures the practical long-term stability of the system.*

Here "practical" refers to times that are very long compared with the expected lifetime of the star [7, §3, footnote 1]. The conjecture is informal: it concerns real systems with actual masses and astronomical time scales, and it is used in [7] to classify observed multi-planet systems.

Chaos and collision safety. The secular dynamics of a planetary system need not be regular, and the AMD-stability criterion does not ask for its stability. It asks only that the eccentricities

and inclinations, however they evolve, never reach a configuration in which orbits touch. In the full Newtonian problem this rests on two facts. First, the total angular momentum L is conserved exactly, not only in an averaged system. In the normalization of Section 2, where the AMD divided by μ is $C = \sum_i \Lambda_i - |L|$, this gives

$$C(t) - C(0) = \sum_{i=1}^{n} \big(\Lambda_i(t) - \Lambda_i(0)\big)\,,$$

so the AMD can change only as much as the Delaunay actions Λ_i, which are functions of the semi-major axes alone. Second, the actions Λ_i are conjugate to the fast angles (the mean longitudes), and averaging over these angles shows that they drift very slowly, for a time that is exponentially long in a negative power of μ. This second fact needs no information about the slow (secular) variables, beyond the requirement that they stay in a region where the Hamiltonian is analytic and the orbits are separated. The AMD, which bounds the eccentricities through $C \geq \sum_i \Lambda_i(1 - \sqrt{1-e_i^2})$, keeps the slow variables in such a region. The two facts therefore close a bootstrap: as long as the axes have moved little, the AMD has moved little, so the orbits are still separated and averaging still applies. The secular motion inside the allowed region can be chaotic.

Results. We make this mechanism rigorous, in the asymptotic regime $\mu \to 0$, up to the strong threshold. Let $B(\Lambda)$ be the least of the pair thresholds $B_{i,i+1}(\Lambda)$ and of the actions Λ_i, computed exactly from the actions $\Lambda = (\Lambda_1, \ldots, \Lambda_n)$; the inequality $C < B(\Lambda)$ is the strong AMD-stability condition of Laskar and Petit (Section 2). Fix the normalized masses $\nu = (\nu_1, \ldots, \nu_n)$, a compact set $\mathcal{A}$ of ordered semi-major axes and a margin $\delta > 0$. Informally, our results are as follows.

- *Theorem A (barrier).* For fixed $m > 0$, every state whose semi-major axes lie in a compact set of ordered axes and whose AMD is at most $B(\Lambda) - m$ has its perihelia bounded below and adjacent orbital annuli separated by a fixed distance, uniformly. No secular or dynamical information enters.

- *Theorems B and C (exponential collision safety).* In the plane and in space, if the initial axes lie in $\mathcal{A}$, the initial AMD is at most $B(\Lambda(0)) - \delta$, and the initial mean motions satisfy a Diophantine condition with constant $\gamma > 0$ and exponent $\tau > n-1$, then the motion exists and is unique for $|t| \leq T_0 \exp(c\mu^{-1/(2(\tau+1))})$. During this time the AMD stays below $B(\Lambda(t)) - \delta/2$, the axes move by $O(\sqrt{\mu})$, and the orbital annuli of the planets remain strictly ordered and uniformly separated from each other and from the star, both for the canonical elements and for the osculating ellipses computed from the actual velocities. No hypothesis on the inclinations is made in space.

- *Corollary D.* For $\tau = n - 1/2$, which includes almost every initial frequency vector, the time is $T_0 \exp(c\mu^{-1/(2n+1)})$.

- *Theorem E (no arithmetic hypothesis).* Without any condition on the frequencies, the same conclusions hold for $|t| \leq T_0 \exp(c\mu^{-\mathfrak{a}_n})$, with drift $O(\mu^{\mathfrak{b}_n})$, where $\mathfrak{a}_n = 2/(6n^2 - 3n + 14)$ and $\mathfrak{b}_n = 3/2\,\mathfrak{a}_n$ are Nekhoroshev's exponents for the planetary problem [8, (12.4)–(12.5)].

In all results the constants depend only on ν, $\mathcal{A}$, δ (and γ, τ), and are chosen before μ and the initial state. The results are uniform over all eccentricities, inclinations and phases allowed by the AMD condition.

Scope. These theorems are a rigorous asymptotic form of the mechanism behind the Laskar–Petit conjecture: for every fixed margin below the strong threshold, collision safety holds for exponentially long times once the masses are small enough. They do not resolve the conjecture itself, which concerns practical time scales and actual masses. Our constants are obtained by compactness and are not explicit, and we make no statement about the Solar System or any observed system. We also prove nothing about the stability of the secular dynamics.

Ideas of the proofs. The barrier (Theorem A) follows from the identity for C, the inequality $C \geq \sum_i \Lambda_i(1 - \sqrt{1 - e_i^2})$ and the continuity of B. For Theorems B and C we work in regular Poincaré variables, in which the eccentricities and inclinations are Cartesian slow coordinates, and C is half the squared norm of the slow variables. We average over the fast angles about the linear Hamiltonian with the initial frequencies, treating the nonlinear Kepler term on an action ball of radius $\sqrt{\mu}$ as part of the perturbation. The point of the construction is that the entire set of slow variables allowed by the AMD condition, a ball, is a single domain of analyticity with a fixed complex width. The normal form is of the type of Pöschel's lemma [11, p. 192]; the Diophantine condition with radius $\sqrt{\mu}$ yields the exponent $1/(2(\tau+1))$, and a first-exit argument closes the bootstrap. Theorem E instead applies Nekhoroshev's theorem with slow variables [8, Theorem 4.4] on finitely many cells covering the sublevel set $\{C \leq B(\Lambda) - \delta\}$.

Relation to earlier work. The mechanism is not new, and Theorem E is close to a classical result. Nekhoroshev's planetary theorem [8, §12, Theorem 12.3] concerns *domains of planetary motions*: the semi-major axes lie in a box, and the total angular momentum of the unperturbed system, divided by μ, has norm at least a constant exceeding the largest value for which the boundary of the domain contains collisions of the planets with one another or with the star [8, §12.2, p. 55]. For initial data in such a domain no collision occurs for a time of order $\mu^{-1}\exp(c\mu^{-\mathfrak{a}_n})$, during which the axes change by $O(\mu^{\mathfrak{b}_n})$ [8, (12.3)–(12.5)]. A lower bound on the angular momentum is an upper bound on the AMD, and the collision value is a strong-AMD threshold taken uniformly over the box. Since B is continuous, covering $\mathcal{A}$ by finitely many small boxes appears to give, from Nekhoroshev's theorem, a statement of the type of Theorem E under the same hypothesis, up to the margin δ. What Theorem E adds is the identification of the hypothesis with the strong threshold of Laskar and Petit, the conclusions on ordered annuli and osculating ellipses, and a formal proof.

We could not consult Niederman's paper [9]; our description of it is taken from [4, 3]. According to [4, §1.3], Niederman extended Nekhoroshev theory to the planetary problem, proving that the actions conjugate to the fast angles are stable over exponentially long times provided the secular actions stay in a bounded region, and that near circular coplanar motions the conservation of angular momentum guarantees this. According to [3, p. 2], his explicit estimates for the planetary three-body problem near a periodic torus give, at the 5:2 resonance, stability for a time comparable with the age of the Solar System if the ratio of the mass of the larger planet to that of the star is at most 10^{-13}. Barbieri and Niederman [3, §3.3.2, Theorem 7] confine the eccentricities through the conservation of angular momentum in a Nekhoroshev estimate for the planar three-body problem that is local near a periodic orbit of the unperturbed system [3, §3.3.3]. Pinzari's construction of Kolmogorov tori in the planetary problem uses an averaging of the fast angles with an exponentially small remainder [10, Proposition 5.1]. In the other direction, Clarke, Féjoz and Guardia construct orbits of the problem with three planets along which the semi-major axis of the outer planet undergoes large variations [4, Theorem 1]; along these orbits the conservation of angular momentum

does not confine the secular variables [4, §1.3].

Theorems B–C and Corollary D differ more substantially from these results. Under the Diophantine condition the drift is $O(\sqrt{\mu})$ and the time exponent is $1/(2(\tau+1))$, compared with $\mu^{\mathfrak{b}_n}$ and $\mathfrak{a}_n$. To our knowledge, these statements have not appeared in this sharp form; since we could not consult [9], this rests in part on the descriptions above. We do not present the normal form as new.

Formal verification. Theorems A–E and Corollary D are proved in the proof assistant Lean 4 with the Mathlib library, in the repository [1] at commit `74d77b43`. The formal statements have the hypotheses, quantifier order and conclusions of Section 2; the dynamical ones also list $\Lambda(0) \in \mathcal{Q}$, which follows from $a(0) \in \mathcal{A}$. More detailed proofs are written out in [2]. Appendix A describes what is verified, the correspondence between statements and formal declarations, and what the formal check does not cover.

Organization. Section 2 sets up the model and states the results. Section 3 proves the barrier and the identities for the angular momentum and the AMD. Section 4 constructs the analytic domain and the normal form and proves Theorem B, and Section 5 treats the spatial case and Corollary D. Section 6 proves Theorem E. Section 7 discusses limitations and open questions, and Appendix A the formalization.

2 Setting and results

2.1 The model

We use units in which the gravitational constant and the mass of the star are equal to one. Fix $n \geq 2$ and $\nu_1, \dots, \nu_n > 0$; the planets have masses $\mu\nu_1, \dots, \mu\nu_n$, where $\mu > 0$ is small. The space dimension is $d \in \{2, 3\}$. Let $q_i \in \mathbb{R}^d$ be the heliocentric position of the i-th planet and $p_i \in \mathbb{R}^d$ its momentum, scaled so that the physical momentum is μp_i. With the symplectic form $\sum_i dp_i \wedge dq_i$, the motion is governed by

$$H_\mu = \sum_{i=1}^{n} \Big(\frac{|p_i|^2}{2\nu_i} - \frac{\nu_i}{|q_i|} \Big) + \mu f \,, \qquad f = \frac{1}{2} \Big| \sum_{i=1}^{n} p_i \Big|^2 - \sum_{1 \leq i < j \leq n} \frac{\nu_i \nu_j}{|q_i - q_j|} \,, \tag{2.1}$$

on the set where $q_i \neq 0$ and $q_i \neq q_j$ for $i \neq j$, in physical time. It is the translation reduction of the Newtonian $(1+n)$-body problem divided by μ, and each solution has a *barycentric reconstruction* solving Newton's equations with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$ (Lemma 3.1).

Elements and the AMD. For each planet let $v_i = p_i/\nu_i$, $E_i = |v_i|^2/2 - 1/|q_i|$ and $\mathbf{e}_i = (|v_i|^2 - 1/|q_i|)q_i - (q_i \cdot v_i)v_i$ be the canonical velocity, the specific energy and the eccentricity vector. When $E_i < 0$, the semi-major axis, the Delaunay action, the eccentricity, the angular momenta and the (normalized) angular momentum deficit are

$$a_i = -\frac{1}{2E_i} \,, \quad \Lambda_i = \nu_i \sqrt{a_i} \,, \quad e_i = |\mathbf{e}_i| \,, \quad L_i = q_i \times p_i \,, \quad L = \sum_{i=1}^{n} L_i \,, \quad C = \sum_{i=1}^{n} \Lambda_i - |L| \,. \tag{2.2}$$

In the plane, $q \times p = q^1 p^2 - q^2 p^1$ is a scalar. If $E_i < 0$, then $e_i \leq 1$ and $r_i^- \leq |q_i| \leq r_i^+$, where $r_i^\pm = a_i(1 \pm e_i)$ are the perihelion and aphelion radii (Lemma 3.2). The physical angular momentum and AMD are μL and μC, so a normalized margin δ is a physical margin $\mu\delta$.

The chamber and the thresholds. Ordered axes correspond to actions in the *chamber*

$$\mathcal{Q} = \left\{\Lambda \in \mathbb{R}^n : \ \Lambda_i > 0, \ a_1(\Lambda) < \cdots < a_n(\Lambda)\right\}, \qquad a_i(\Lambda) = (\Lambda_i/\nu_i)^2 . \tag{2.3}$$

Let $D(u) = 1 - \sqrt{1-u^2}$ for $u \in [0,1]$. For $\Lambda \in \mathcal{Q}$ and $i < j$, define

$$\begin{aligned} B_{ij}(\Lambda) &= \min\left\{\Lambda_i D(u) + \Lambda_j D(v) : \ u, v \in [0,1], \ a_i(\Lambda)(1+u) \geq a_j(\Lambda)(1-v)\right\}, \\ B_{\mathrm{cross}}(\Lambda) &= \min_{1 \leq i < n} B_{i,i+1}(\Lambda), \qquad B_\star(\Lambda) = \min_{1 \leq i \leq n} \Lambda_i, \qquad B(\Lambda) = \min\left(B_{\mathrm{cross}}(\Lambda), B_\star(\Lambda)\right). \end{aligned} \tag{2.4}$$

The minima are attained and B is positive and continuous on $\mathcal{Q}$ (Lemma 3.4). Here $B_{ij}(\Lambda)$ is the least deficit for which the annuli of planets i and j touch, and Λ_i that of a radial orbit. Since $C \geq \sum_i \Lambda_i D(e_i)$ (Lemma 3.3), the inequality $C < B(\Lambda)$ excludes contact of adjacent annuli and $e_i = 1$. After division by Λ_j, B_{ij} is the critical AMD of [7, §3.2], and $C < B(\Lambda)$ is the strong AMD-stability of [7, §5 and §5.3] (Lemma 3.4).

Frequencies and data. The Kepler Hamiltonian and its mean motions are

$$h(\Lambda) = -\sum_{i=1}^{n} \frac{\nu_i^3}{2\Lambda_i^2}, \qquad \omega_i(\Lambda) = \partial_{\Lambda_i} h(\Lambda) = \frac{\nu_i^3}{\Lambda_i^3} = a_i(\Lambda)^{-3/2} . \tag{2.5}$$

For $\gamma > 0$ and $\tau > n-1$, a vector Λ^0 is (γ,τ)-*Diophantine* if

$$|k \cdot \omega(\Lambda^0)| \geq \gamma\, |k|_1^{-\tau} \qquad \text{for every } k \in \mathbb{Z}^n \setminus \{0\}, \qquad |k|_1 = \sum_{i=1}^{n} |k_i| . \tag{2.6}$$

Throughout, $\mathcal{A} \subset \{a \in \mathbb{R}^n : 0 < a_1 < \cdots < a_n\}$ is compact and $\delta > 0$. An initial state is *admissible* if $q_i(0) \neq 0$ and $E_i(0) < 0$ for every i, and

$$a(0) \in \mathcal{A}, \qquad C(0) \leq B(\Lambda(0)) - \delta . \tag{2.7}$$

Here $a(0) \in \mathcal{A}$ implies $\Lambda(0) \in \mathcal{Q}$, so $B(\Lambda(0))$ is defined. For $T_0, c > 0$ we put

$$T_\mu = T_0 \exp\left(c\, \mu^{-\frac{1}{2(\tau+1)}}\right), \tag{2.8}$$

and we write $\|x\|_\infty = \max_i |x_i|$ for vectors of actions or axes.

Osculating elements. The actual heliocentric velocity is $V_i = \dot{q}_i = p_i/\nu_i + \mu \sum_j p_j$, not v_i. With $k_i = 1 + \mu\nu_i$, the velocity-osculating elements are

$$E_i^{\mathrm{osc}} = \frac{|V_i|^2}{2} - \frac{k_i}{|q_i|}, \quad a_i^{\mathrm{osc}} = -\frac{k_i}{2E_i^{\mathrm{osc}}}, \quad \mathbf{e}_i^{\mathrm{osc}} = \frac{1}{k_i}\left(\left(|V_i|^2 - \frac{k_i}{|q_i|}\right) q_i - (q_i \cdot V_i)\, V_i\right), \quad e_i^{\mathrm{osc}} = |\mathbf{e}_i^{\mathrm{osc}}| ,$$

and $r_i^{\mathrm{osc},\pm} = a_i^{\mathrm{osc}}(1 \pm e_i^{\mathrm{osc}})$. When $E_i^{\mathrm{osc}} < 0$ they describe the Kepler ellipse through q_i tangent to V_i, the orbit computed from the actual position and velocity (Lemma 4.7).

2.2 Results

Theorem A (Strong AMD barrier)**.** *Let $d = 2$, $n \geq 2$, $\nu \in (0,\infty)^n$, let $U \subset \mathcal{Q}$ be compact and $m > 0$. There exist $d_\star, d_{\mathrm{pair}} > 0$ and a compact set K of states with $q_i \neq 0$, $q_i \neq q_j$ $(i \neq j)$ and $E_i < 0$ for all i, such that:*

(i) if $\Lambda \in U$, $e \in [0,1]^n$ and $\sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m$, then $a_i(\Lambda)(1-e_i) \geq d_\star$ for $1 \leq i \leq n$ and $a_{i+1}(\Lambda)(1-e_{i+1}) - a_i(\Lambda)(1+e_i) \geq d_{\mathrm{pair}}$ for $1 \leq i < n$;

(ii) every state with $q_i \neq 0$ and $E_i < 0$ for all i, $\Lambda \in U$ and $C \leq B(\Lambda) - m$ belongs to K and satisfies $r_i^- \geq d_\star$, $|q_i| \geq d_\star$, $|q_i - q_j| \geq d_{\mathrm{pair}}$, $|L_i| \geq m$ and $L_i L > 0$ for all $i \neq j$, and $r_{i+1}^- - r_i^+ \geq d_{\mathrm{pair}}$ for $i < n$.

Theorem B (Planar exponential collision safety)**.** *Let $d = 2$, $n \geq 2$, $\nu \in (0,\infty)^n$, $\mathcal{A}$ and δ as above, $\gamma > 0$ and $\tau > n - 1$. There exist positive constants $\mu_0, c, T_0, A_\Lambda, A_a, d_\star, d_{\mathrm{pair}}, A_{\mathrm{osc}}, A_{\mathrm{el}}$, depending only on $(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$ and satisfying $A_{\mathrm{osc}}\mu_0 \leq d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \leq d_{\mathrm{pair}}/2$, such that the following holds for every $\mu \in (0, \mu_0]$ and every admissible initial state with (γ,τ)-Diophantine $\Lambda(0)$. Let $T = T_\mu$.*

(i) Well-posedness. *There is a solution of (2.1) on $[-T,T]$ with the given initial state. Every solution with the same initial state on an open interval containing 0 agrees with it on the common part of $[-T,T]$. Its barycentric reconstruction solves Newton's equations with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$ on $[-T,T]$.*

Moreover, for every $t \in [-T,T]$ and all $i \neq j$, with all elements evaluated at time t and the inequalities involving $i+1$ required for $i < n$:

(ii) AMD margin. *$q_i \neq 0$, $E_i < 0$, $\Lambda(t) \in \mathcal{Q}$ and $C(t) \leq B(\Lambda(t)) - \delta/2$.*

(iii) Drift. *$\|\Lambda(t) - \Lambda(0)\|_\infty \leq A_\Lambda\sqrt{\mu}$, $\|a(t) - a(0)\|_\infty \leq A_a\sqrt{\mu}$ and $|C(t) - C(0)| \leq nA_\Lambda\sqrt{\mu}$.*

(iv) Separation. *$r_i^- \geq d_\star$, $r_{i+1}^- - r_i^+ \geq d_{\mathrm{pair}}$, $|q_i| \geq d_\star$ and $|q_i - q_j| \geq d_{\mathrm{pair}}$.*

(v) Osculating orbits. *$E_i^{\mathrm{osc}} < 0$, $a_i^{\mathrm{osc}} > 0$, $e_i^{\mathrm{osc}} < 1$, $r_i^{\mathrm{osc},-} \leq |q_i| \leq r_i^{\mathrm{osc},+}$, and*

$$|a_i^{\mathrm{osc}} - a_i| + |\mathbf{e}_i^{\mathrm{osc}} - \mathbf{e}_i| \leq A_{\mathrm{el}}\mu\,, \quad |r_i^{\mathrm{osc},\pm} - r_i^\pm| \leq A_{\mathrm{osc}}\mu\,, \quad r_i^{\mathrm{osc},-} \geq \frac{d_\star}{2}\,, \quad r_{i+1}^{\mathrm{osc},-} - r_i^{\mathrm{osc},+} \geq \frac{d_{\mathrm{pair}}}{2}\,.$$

Theorem C (Spatial exponential collision safety)**.** *For $d = 3$, Theorem B holds verbatim, with the same data and quantifiers and the conclusions* (i)–(v)*, where $|\cdot|$ is the Euclidean norm on $\mathbb{R}^3$ and L is the vector total angular momentum.*

Corollary D (Exponent $1/(2n+1)$)**.** *Theorems B and C hold with $\tau = n - 1/2$, and then $T_\mu = T_0 \exp(c\mu^{-1/(2n+1)})$.*

Theorem E (Collision safety without arithmetic hypothesis)**.** *Let $d \in \{2,3\}$, $n \geq 2$, $\nu \in (0,\infty)^n$, $\mathcal{A}$ and δ as above, and put $\mathfrak{a}_n = 2/(6n^2 - 3n + 14)$ and $\mathfrak{b}_n = 3/(6n^2 - 3n + 14)$. There exist positive constants $\mu_0, c, T_0, A_\Lambda, A_a, d_\star, d_{\mathrm{pair}}, A_{\mathrm{osc}}, A_{\mathrm{el}}$, depending only on $(n, \nu, \mathcal{A}, \delta)$ and satisfying $A_{\mathrm{osc}}\mu_0 \leq d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \leq d_{\mathrm{pair}}/2$, such that for every $\mu \in (0, \mu_0]$ and every admissible initial state, with $T = T_0 \exp(c\mu^{-\mathfrak{a}_n})$, the conclusions* (i), (ii), (iv) *and* (v) *of Theorem B hold, and* (iii) *holds with $\sqrt{\mu}$ replaced by $\mu^{\mathfrak{b}_n}$.*

2.3 Comments

Constants. All constants are chosen from the data, before μ and the initial state, and are uniform over all phases, eccentricities, inclinations and orientations allowed by (2.7). The proofs may assume

$\max_i \nu_i = 1$: with $M = \max_i \nu_i$, the substitution $(\mu, \nu, p) \mapsto (M\mu, \nu/M, p/M)$ preserves the masses, the motion, the axes and the frequencies, and divides Λ, C, B by M. The theorems concern the limit $\mu \to 0$ with ν, $\mathcal{A}$, δ fixed, and assume nothing about secular stability. Theorems B and E do not imply each other (Table 1).

No orbit crossing. By (iv), the entire annuli $\{r_i^- \le |x| \le r_i^+\}$ stay ordered and d_{pair}-apart, and $d_\star$ from the star; this is stronger than separation of the positions. By (v) the same holds for the osculating ellipses with half the distances.

Space. Theorem C assumes nothing on inclinations or orientations; $C < B_\star$ already forces $L \neq 0$ and $L_i \cdot L > 0$ (Lemma 3.3). The proof of Theorem A does not use the dimension; for $d = 3$ it gives the same conclusions with $L_i \cdot L \ge m|L|$ in place of $|L_i| \ge m$, $L_i L > 0$, as used for Theorem C.

3 The strong barrier

This section is elementary and contains no small parameter. The inequality $C \ge \sum_i \Lambda_i D(e_i)$ turns the strong condition $C \le B(\Lambda) - m$ into a bound on the eccentricities that keeps every perihelion and every pair of adjacent radial annuli uniformly apart, and B is the least deficit at which two adjacent annuli can touch or a planet can reach the star. Conservation of L then makes C move only with the actions (Lemma 3.6). Throughout, $d \in \{2, 3\}$, and in the plane $q \times p$ is the oriented scalar determinant.

Lemma 3.1 (Normalization)**.** *For every $d \in \{2, 3\}$ and all $\nu_i, \mu > 0$, the Hamiltonian (2.1) is the translation reduction, at zero total momentum, of the Newtonian $(1+n)$-body Hamiltonian with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$, divided by μ, with unchanged time. Its solutions reconstruct Newtonian solutions with these masses.*

Proof. In barycentric coordinates put $P_i = \mu p_i$, $P_0 = -\sum_i P_i$ and $q_i = x_i - x_0$, so that $\sum_{i=0}^n P_i \cdot dx_i = \sum_{i=1}^n P_i \cdot dq_i$. The kinetic energy is $\sum_i |P_i|^2/(2\mu\nu_i) + \frac{1}{2}|\sum_i P_i|^2$ and the potential $-\sum_i \mu\nu_i/|q_i| - \sum_{i<j} \mu^2\nu_i\nu_j/|q_i - q_j|$; the symplectic form and the Hamiltonian both acquire the factor μ, which cancels and gives (2.1). Conversely, put $x_0 = -\mu \sum_i \nu_i q_i/(1 + \mu\sum_i \nu_i)$ and $x_i = x_0 + q_i$; Hamilton's equations give $\dot{x}_0 = -\mu \sum_j p_j$, $\dot{x}_i = p_i/\nu_i$, and Newton's equations for the momenta μp_i. □

Lemma 3.2 (Orbital identities)**.** *Let $k > 0$, $q \neq 0$, $v \in \mathbb{R}^d$ with $E = |v|^2/2 - k/|q| < 0$, and put $a = -k/(2E)$, $\mathbf{e} = \big((|v|^2 - k/|q|)q - (q \cdot v)v\big)/k$ and $e = |\mathbf{e}|$. Then, including the radial case $e = 1$,*

$$e^2 = 1 + \frac{2E|q \times v|^2}{k^2} = 1 - \frac{|q \times v|^2}{ka} \le 1, \qquad a(1-e) \le |q| \le a(1+e), \qquad |v|^2 = k\Big(\frac{2}{|q|} - \frac{1}{a}\Big). \tag{3.1}$$

For $k = 1$ and $v = p/\nu$ these are the elements (2.2), and $|q \times p| = \Lambda\sqrt{1 - e^2}$.

Proof. Expanding $k^2 e^2$ and using $|q \times v|^2 = |q|^2|v|^2 - (q \cdot v)^2$ gives the first identity; substitute $2E = -k/a$ for the second. With $r = |q|$, the energy identity gives $(q \cdot v)^2 = 2kr - kr^2/a - |q \times v|^2 \ge 0$; inserting $|q \times v|^2 = ka(1 - e^2)$ and completing the square gives $(r - a)^2 \le a^2 e^2$. The last identity is the definition of E, and $|q \times p| = \nu|q \times v| = \nu\sqrt{a(1 - e^2)}$ for $k = 1$. □

Lemma 3.3 (Angular momentum and the AMD identity)**.** *Along every solution of (2.1) on an interval, L is constant. In the negative-energy region $C \geq \sum_i \Lambda_i D(e_i) \geq 0$. If $C < B_\star(\Lambda)$, then $L \neq 0$, the projections $H_i = L_i \cdot L/|L|$ are positive, and*

$$C(t) - C(0) = \sum_i \big(\Lambda_i(t) - \Lambda_i(0)\big), \qquad C = \sum_i (\Lambda_i - H_i). \tag{3.2}$$

If $C \leq B_\star(\Lambda) - m$ with $m > 0$, then $H_i \geq m$ for every i. In the plane, after choosing the orientation with $L > 0$, every L_i is positive and $H_i = L_i$.

Proof. In $\frac{d}{dt}\sum_i q_i \times p_i$ the kinetic terms $(p_i/\nu_i) \times p_i$ and $\mu(\sum_j p_j) \times (\sum_i p_i)$ vanish, central forces have zero torque, and the two forces of a pair give a multiple of $(q_i - q_j) \times (q_i - q_j) = 0$; so $\dot{L} = 0$. By (3.1), $|L| \leq \sum_i |L_i| = \sum_i \Lambda_i \sqrt{1 - e_i^2}$, which gives $C \geq \sum_i \Lambda_i D(e_i) \geq 0$. If $L = 0$, then $C = \sum_i \Lambda_i \geq B_\star$. Otherwise $|L| = \sum_i H_i$ gives the second identity in (3.2), and constancy of $|L|$ gives the first. Since each $\Lambda_i - H_i \geq \Lambda_i - |L_i| \geq 0$, we get $H_i \geq \Lambda_i - C$, which is positive if $C < B_\star$ and at least m if $C \leq B_\star - m$. □

Lemma 3.4 (The thresholds)**.** *For all $\nu_i > 0$ and $\Lambda \in \mathcal{Q}$, every minimum in (2.4) is attained and positive, and B_{ij} and B are continuous on $\mathcal{Q}$. The minimum defining B_{ij} is attained at radial contact $a_i(1 + u) = a_j(1 - v)$, and*

$$B_{ij}(\Lambda) < \Lambda_j, \qquad B(\Lambda) = \min\big(B_{\text{cross}}(\Lambda), \Lambda_1\big). \tag{3.3}$$

With $\alpha = a_i/a_j$, the quotient $B_{ij}(\Lambda)/\Lambda_j$ is the relative critical AMD $C_c(\alpha, \nu_i/\nu_j)$ of Laskar and Petit [7, §§3.1–3.2].

Proof. Fix $i < j$, so $0 < \alpha < 1$. The feasible set $\{(u, v) \in [0, 1]^2 : \alpha u + v \geq 1 - \alpha\}$ is compact and contains $(0, 1 - \alpha)$, and the objective $\Lambda_i D(u) + \Lambda_j D(v)$ is continuous, so the minimum is attained. If (u, v) is feasible, then $t = (1 - \alpha)/(\alpha u + v) \leq 1$, the point (tu, tv) lies on the contact line $\alpha u + v = 1 - \alpha$, and since D is increasing the cost does not increase. Hence the minimum is attained at contact, which we parametrize over a fixed interval:

$$u = \bar{u}_\alpha t, \qquad v = 1 - \alpha - \alpha \bar{u}_\alpha t, \qquad t \in [0, 1], \qquad \bar{u}_\alpha = \min\Big(1, \frac{1 - \alpha}{\alpha}\Big). \tag{3.4}$$

The resulting cost $F(\Lambda, t)$ is continuous, and $|\min_t F(\Lambda, t) - \min_t F(\Lambda', t)| \leq \sup_t |F(\Lambda, t) - F(\Lambda', t)|$ with uniform continuity on compact sets proves continuity of B_{ij}, hence of B. A zero cost forces $u = v = 0$, which is not feasible. The competitor $(0, 1 - \alpha)$ gives $B_{ij} \leq \Lambda_j D(1 - \alpha) < \Lambda_j$. In particular $\Lambda_i > B_{i-1,i} \geq B_{\text{cross}}$ for $i \geq 2$, which gives the second identity in (3.3).

Finally $\Lambda_i/\Lambda_j = (\nu_i/\nu_j)\sqrt{\alpha}$, so B_{ij}/Λ_j is the minimum of $(\nu_i/\nu_j)\sqrt{\alpha}\, D(u) + D(v)$ on the contact line $\alpha u + v = 1 - \alpha$, $u, v \in [0, 1]$. This is the minimization of the relative AMD under the collision condition that defines C_c in [7, §§3.1–3.2], with mass ratio ν_i/ν_j. □

The threshold is geometrically sharp. If (u, v) minimizes B_{ij}, two coplanar prograde ellipses with eccentricities u, v and opposite perihelion directions touch, the inner aphelion meeting the outer perihelion, and their combined deficit is B_{ij}; a radial orbit has deficit Λ_i and is the limit of prograde ellipses as $e_i \uparrow 1$. This is not a claim that motion above the threshold must collide.

Lemma 3.5 (Uniform barrier)**.** *Let $U \subset \mathcal{Q}$ be compact and $m > 0$. There exist $d_\star, d_{\text{pair}} > 0$, depending only on (n, ν, U, m), such that for every $\Lambda \in U$ and $e \in [0, 1]^n$ with $\sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m$,*

$$a_i(\Lambda)(1 - e_i) \geq d_\star, \qquad a_{i+1}(\Lambda)(1 - e_{i+1}) - a_i(\Lambda)(1 + e_i) \geq d_{\text{pair}}. \tag{3.5}$$

Every state with $q_i \neq 0$, $E_i < 0$, $\Lambda \in U$ and $C \leq B(\Lambda) - m$ satisfies (3.5), $|q_i| \geq d_\star$ and $|q_i - q_j| \geq d_{\mathrm{pair}}$ for $i \neq j$, and $H_i \geq m$; these states form a compact subset of the region $\{q_i \neq 0,\ q_i \neq q_j,\ E_i < 0\}$.

Proof. Let $K_m = \{(\Lambda, e) \in U \times [0,1]^n : \sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m\}$, which is compact by Lemma 3.4. If an adjacent separation in (3.5) were nonpositive at a point of K_m, then (e_i, e_{i+1}) would be feasible for $B_{i,i+1}$ and $\sum_k \Lambda_k D(e_k) \geq B_{i,i+1}(\Lambda) \geq B(\Lambda)$, a contradiction. So the minimum d_{pair} of the continuous separations over K_m is positive (if $K_m = \emptyset$ all is vacuous). Put

$$\Lambda_{\max} = \max_{\Lambda \in U,\, i} \Lambda_i, \qquad a_{\min} = \min_{\Lambda \in U,\, i} a_i(\Lambda), \qquad a_{\max} = \max_{\Lambda \in U,\, i} a_i(\Lambda), \qquad d_\star = \frac{a_{\min} m^2}{2\Lambda_{\max}^2}. \tag{3.6}$$

As $B \leq B_\star \leq \Lambda_i$, the deficit bound gives $\Lambda_i D(e_i) \leq \Lambda_i - m$, i.e. $\sqrt{1 - e_i^2} \geq m/\Lambda_{\max}$; with $1 - e_i \geq (1 - e_i^2)/2$ this gives the first bound in (3.5).

For a physical state, Lemma 3.3 gives $\sum_i \Lambda_i D(e_i) \leq C \leq B(\Lambda) - m$, so (3.5) holds, and $H_i \geq m$ because $B \leq B_\star$. By (3.1), $d_\star \leq |q_i| \leq 2a_{\max}$, $|p_i| \leq \nu_i \sqrt{2/d_\star}$ and $E_i \leq -1/(2a_{\max})$, and summing adjacent separations gives $|q_i - q_j| \geq |q_j| - |q_i| \geq d_{\mathrm{pair}}$ for $i < j$. These inequalities define a compact set inside the nonsingular region, on which $\Lambda \in U$ and $C \leq B(\Lambda) - m$ are closed conditions. □

Theorem A is Lemma 3.5: in the plane, with the orientation $L > 0$ of Lemma 3.3, $H_i = L_i = |L_i| \geq m$. The next lemma converts action drift into margin drift.

Lemma 3.6 (Moving threshold and continuation)**.** *Let $\mathcal{L} \subset \mathcal{Q}$ be compact and $\delta > 0$. There exist a radius $r_g > 0$ and a compact $U \subset \mathcal{Q}$ containing the closed $\|\cdot\|_\infty$-neighbourhood of radius $4r_g$ of $\mathcal{L}$ such that the modulus*

$$\Omega_B(r) = \sup\{|B(\Lambda) - B(\Lambda')| : \Lambda, \Lambda' \in U,\ \|\Lambda - \Lambda'\|_\infty \leq r\}$$

satisfies $\Omega_B(r) \to 0$ as $r \downarrow 0$ and $\Omega_B(4r_g) \leq \delta/16$. Let $0 < r \leq 4r_g$. If a solution of (2.1) has $\Lambda^0 = \Lambda(0) \in \mathcal{L}$, $C(0) \leq B(\Lambda^0) - \delta$, and $\|\Lambda(t) - \Lambda^0\|_\infty \leq r$ while $E_i(t) < 0$, then

$$C(t) \leq B(\Lambda(t)) - \delta + nr + \Omega_B(r). \tag{3.7}$$

If $nr + \Omega_B(r) \leq \delta/2$, the maximal solution cannot end, forwards or backwards, at a finite time while these conditions hold.

Proof. Take $\rho > 0$ with the closed 2ρ-neighbourhood U of $\mathcal{L}$ inside $\mathcal{Q}$. By Lemma 3.4, B is uniformly continuous on U, so $\Omega_B(r) \to 0$; choose $4r_g \leq \rho$ with $\Omega_B(4r_g) \leq \delta/16$. Lemma 3.3 gives $C(t) = C(0) + \sum_i (\Lambda_i(t) - \Lambda_i^0) \leq B(\Lambda^0) - \delta + nr$, and $B(\Lambda^0) \leq B(\Lambda(t)) + \Omega_B(r)$ gives (3.7). If $nr + \Omega_B(r) \leq \delta/2$, then $C(t) \leq B(\Lambda(t)) - \delta/2$, and Lemma 3.5 with $m = \delta/2$ keeps the state in a fixed compact subset of the domain of the analytic vector field of (2.1), which a maximal solution must leave at a finite endpoint, in either time direction. □

4 The planar theorem

By Section 3, Theorem B reduces to the actions: if $\Lambda(t)$ stays within $O(\sqrt{\mu})$ of Λ^0 for $|t| \leq T_\mu$, then (3.7) keeps the AMD $\delta/2$ below the moving threshold, and Lemma 3.5 gives every separation. We control the actions by averaging over the fast angles, with three features.

First, we average about the *linear* Hamiltonian $h_0 = h(\Lambda^0) + \omega(\Lambda^0) \cdot (\Lambda - \Lambda^0)$ with the initial Kepler frequencies; the Kepler nonlinearity $h - h_0$ is part of the perturbation on an action tube of

radius r. Its vector field has size r, that of μf has size μ/r in the action directions, and $r=\sqrt{\mu}$ balances the two. Second, the secular variables $z=(x_i,y_i)_i$ are carried along as canonical variables, with no assumption on their dynamics; the averaged Hamiltonian is never analysed. Third, the slow domain is the entire ball of secular states allowed by the AMD margin; all its configurations are collision free with a fixed margin, so it is one analytic domain with fixed complex widths. Since the AMD equals $\frac{1}{2}|z|^2$, the orbit cannot leave the ball before its actions drift or its margin is lost, and a first-exit argument closes the bootstrap.

4.1 The chart and its analytic domain

We use the regular Poincaré variables. For one body with parameter $\nu>0$, given $\Lambda>0$ and $(x,y)\in\mathbb{R}^2$ with $\Gamma=(x^2+y^2)/2<\Lambda$, put

$$a=(\Lambda/\nu)^2,\qquad \beta=1-\frac{\Gamma}{\Lambda},\qquad \varrho=\sqrt{\frac{1-\Gamma/(2\Lambda)}{\Lambda}},\qquad k=\varrho x,\qquad l=-\varrho y,$$

so that $k^2+l^2=1-\beta^2<1$. For $\lambda\in\mathbb{R}$ the eccentric longitude F is the unique solution of

$$F-k\sin F+l\cos F=\lambda, \tag{4.1}$$

and with $c(F)=(\cos F,\sin F)$, $c'(F)=(-\sin F,\cos F)$ and $D_F=1-k\cos F-l\sin F$ we set

$$\mathbf{M}=\begin{pmatrix}1-\frac{l^2}{1+\beta} & \frac{kl}{1+\beta}\\ \frac{kl}{1+\beta} & 1-\frac{k^2}{1+\beta}\end{pmatrix},\qquad q=a\big(\mathbf{M}c(F)-(k,l)\big),\qquad p=\frac{\nu}{\sqrt{a}}\,\frac{\mathbf{M}c'(F)}{D_F}. \tag{4.2}$$

Lemma 4.1 (Regular planar chart)**.** *For every $\nu>0$, the map $(\Lambda,\lambda,x,y)\mapsto(q,p)$ of (4.1)–(4.2), with λ read modulo 2π, is a real analytic diffeomorphism from $\{\Lambda>0,\ \lambda\in\mathbb{T},\ (x,y)\in\mathbb{R}^2,\ \Gamma<\Lambda\}$ onto the planar states with $q\neq0$, $E<0$ and $q\times p>0$. It carries $dp\wedge dq$ to $d\Lambda\wedge d\lambda+dx\wedge dy$; in it the Kepler energy $|p|^2/(2\nu)-\nu/|q|$ is $-\nu^3/(2\Lambda^2)$, the angular momentum is $\Lambda-\Gamma$, and the eccentricity vector is (k,l). It is regular at circular orbits.*

Sketch. The left side of (4.1) has derivative $D_F\geq1-\sqrt{k^2+l^2}>0$ and increases by 2π per period, so F exists, is unique and is analytic. Where $e=\sqrt{k^2+l^2}>0$, write $(k,l)=e(\cos\varpi,\sin\varpi)$ and $F=\psi+\varpi$. In the frame rotated by ϖ, (4.2) is the classical parametrization by the eccentric anomaly ψ, with $\beta=\sqrt{1-e^2}$; it gives the energy, the angular momentum $G=\Lambda\beta=\Lambda-\Gamma$ and the eccentricity vector, also at $e=0$ by continuity. With $\ell=\psi-e\sin\psi$, so that $\lambda=\ell+\varpi$, differentiation and the contribution $G\,d\varpi$ of the rotation give

$$p\cdot dq=\Lambda\,d\ell+G\,d\varpi+d(2\Lambda e\sin\psi)=\Lambda\,d\lambda+\tfrac{1}{2}(x\,dy-y\,dx)+d(2\Lambda e\sin\psi),$$

because $(x,y)=\sqrt{2\Gamma}(\cos\varpi,-\sin\varpi)$ gives $\Gamma\,d(-\varpi)=\frac{1}{2}(x\,dy-y\,dx)$. Exterior differentiation proves the symplectic identity for $e>0$, and analyticity extends it to $e=0$. The inverse recovers Λ from the energy, Γ from the angular momentum, (k,l) as the eccentricity vector, $c(F)=\mathbf{M}^{-1}(q/a+(k,l))$, and $e^{i\lambda}=e^{iF}\exp(-i(k\sin F-l\cos F))$, analytically and without a perihelion angle; see [2]. □

For n planets we use the product chart, with $z=(x_i,y_i)_{i\leq n}\in\mathbb{R}^{2n}$. It covers the states with $q_i\neq0$, $E_i<0$ and $L_i>0$ for all i; on them $H_\mu=h(\Lambda)+\mu f(\Lambda,\lambda,z)$ exactly, f being the pulled-back perturbation of (2.1), and if moreover $L>0$,

$$C=\sum_i\Lambda_i-\sum_i L_i=\sum_i\Gamma_i=\tfrac{1}{2}|z|^2. \tag{4.3}$$

Let $\mathcal{L} = \{(\nu_i\sqrt{a_i})_i : a \in \mathcal{A}\} \subset \mathcal{Q}$ and $\mathcal{L}_\delta = \{\Lambda^0 \in \mathcal{L} : B(\Lambda^0) \geq \delta\}$. Since $C \geq 0$, every initial state satisfying (2.7) has $\Lambda^0 \in \mathcal{L}_\delta$. We fix r_g, U, Ω_B by Lemma 3.6 applied to $\mathcal{L}_\delta$: $4r_g$ is a fixed action radius over which B varies by at most $\delta/16$, and U contains all actions within $4r_g$ of $\mathcal{L}_\delta$. For complex arguments, $|z|^2$ and Γ_i are bilinear sums of squares, and distances are in the maximum norm.

Lemma 4.2 (Analytic domain of the whole sublevel ball)**.** *There are $r_* \in (0, 4r_g]$ and $s, b, M > 0$, depending only on $(n, \nu, \mathcal{A}, \delta)$, such that for every $\Lambda^0 \in \mathcal{L}_\delta$, with*

$$\mathcal{Z}_0(\Lambda^0) = \left\{z \in \mathbb{R}^{2n} : \tfrac{1}{2}|z|^2 \leq B(\Lambda^0) - \delta/4\right\}, \tag{4.4}$$

the product chart and f are holomorphic, 2π-periodic in λ and real on real points in the complex domain

$$\|\Lambda - \Lambda^0\|_\infty < r_*, \qquad \|\operatorname{Im}\lambda\|_\infty < 4s, \qquad \operatorname{dist}_\infty\left(z, \mathcal{Z}_0(\Lambda^0)\right) < 4b, \tag{4.5}$$

and on a slightly smaller domain of the same form the chart, the momenta, f and their first derivatives are bounded by M. Every Newtonian denominator $|q_i|$, $|q_i - q_j|$ has a single holomorphic branch there, with modulus bounded below by a positive constant depending only on $(n, \nu, \mathcal{A}, \delta)$.

Proof. We may assume $\mathcal{L}_\delta \neq \emptyset$; it is compact.

Step 1: every real point of the ball is safe. Let $\Lambda^0 \in \mathcal{L}_\delta$, Λ real with $\|\Lambda - \Lambda^0\|_\infty \leq 4r_g$ (so $\Lambda \in U$), and $z \in \mathcal{Z}_0(\Lambda^0)$. As $\Omega_B(4r_g) \leq \delta/16$,

$$\sum_i \Gamma_i = \tfrac{1}{2}|z|^2 \leq B(\Lambda^0) - \frac{\delta}{4} \leq B(\Lambda) - \frac{3\delta}{16}.$$

Since $B(\Lambda) \leq \Lambda_i$, we get $\Gamma_i \leq \Lambda_i - 3\delta/16$: the point lies in the chart domain with a fixed margin, and $e_i \leq e_* < 1$ for a fixed e_*. At every phase λ the physical state has all $L_i > 0$ and, by (4.3), $C \leq B(\Lambda) - 3\delta/16$, so Lemma 3.5 with $m = 3\delta/16$ gives $d_0 \leq |q_i|, |q_i - q_j| \leq Q$ with fixed $d_0, Q > 0$. The set K' of all such (Λ, λ, z), over $\Lambda^0 \in \mathcal{L}_\delta$ and λ in a closed period cell, is compact.

Step 2: a uniform complex neighbourhood. Since $D_F \geq 1 - e_*$, the Newton map for (4.1) contracts a complex disk of fixed radius about each real root, giving a bounded holomorphic branch of F; the radicands and $1 + \beta$ stay away from zero. By the identity theorem, local extensions agree on overlaps, and a finite subcover of K' gives one complex neighbourhood on which the chart is single valued and periodic. For a denominator, if the complex vector w lies within Euclidean distance η of a real w_0 with $d_0 \leq |w_0| \leq Q$, then

$$|w \cdot w - |w_0|^2| \leq 2Q\eta + \eta^2 \leq d_0^2/2$$

for small fixed η, so $\operatorname{Re}(w \cdot w) \geq d_0^2/2$, the root with positive real part is holomorphic, and its reciprocal is at most $\sqrt{2}/d_0$. Cauchy's estimates bound first derivatives. Finally choose $r_* \leq 4r_g$ and s, b so small that (4.5), which lies within distance $\max(r_*, 4s, 4b)$ of K', and a slight enlargement of it lie in this neighbourhood. □

Thus the slow domain is the *entire* ball (4.4), with complex widths independent of Λ^0 and μ. The proof uses only the safety of every real point of the ball, compactness and the bilinear bound on the denominators, so it applies verbatim to any analytic chart with the identity (4.3).

4.2 Exponential averaging with evolving slow variables

The next three lemmas are standard normal-form estimates of Nekhoroshev and Pöschel type, close to Pöschel's Normal Form Lemma [11, p. 192] (see also [8]); we do not claim them as new, and we only sketch their proofs. More detailed proofs, with constants, are in the supplementary proof notes [2] (Lemmas 9–11 there), where the standard flow estimates of the Lie step are also only sketched. We use a norm on Hamiltonian vector fields, a linear unperturbed Hamiltonian, and n_s slow pairs near an arbitrary compact set; the slow variables enter only through Cauchy estimates, which need nothing of the shape of that set.

Fix $n \geq 1$, $n_s \geq 0$, a nonempty compact $S \subset \mathbb{R}^{2n_s}$, $\Lambda^0 \in \mathbb{R}^n$ and widths $r, s, b > 0$, use the symplectic form $d\Lambda \wedge d\lambda + \sum dx \wedge dy$, and put $\mathsf{D} = 2n + 2n_s$. For $1 \leq \rho \leq 4$ let

$$\mathcal{D}_\rho = \big\{(\Lambda, \lambda, z) : \|\Lambda - \Lambda^0\|_\infty < \rho r,\ \|\mathrm{Im}\,\lambda\|_\infty < \rho s,\ \mathrm{dist}_\infty(z, S) < \rho b\big\}. \tag{4.6}$$

For a λ-periodic holomorphic vector field V with scaled components $\widehat{V} = (V_\Lambda/r, V_\lambda/s, V_z/b)$ and Fourier coefficients $\widehat{V}_k(\Lambda, z)$, let $\|V\|_\rho = \sum_{k\in\mathbb{Z}^n} e^{\rho s|k|_1} \sup|\widehat{V}_k|_\infty$, the supremum over $\|\Lambda - \Lambda^0\|_\infty < \rho r$, $\mathrm{dist}_\infty(z, S) < \rho b$.

Lemma 4.3 (Fourier and commutator estimates)**.** *For $0 < \sigma < \rho - 1$ the norm decreases under restriction, the projections $V \mapsto \langle V\rangle$ and $V \mapsto V - \langle V\rangle$ onto the λ-average and its complement do not increase $\|\cdot\|_\rho$, and*

$$\|[V, W]\|_{\rho-\sigma} \leq \frac{2\mathsf{D}}{\sigma}\|V\|_\rho\|W\|_\rho. \tag{4.7}$$

If $|\widehat{V}|_\infty \leq M'$ on $\mathcal{D}_4$, then $\|V\|_3 \leq M'\big((1 + e^{-s})/(1 - e^{-s})\big)^n$.

Sketch. Contour shifts give $|\widehat{V}_k| \leq M'e^{-4s|k|_1}$. A disk of radius σr or σb about a point of the smaller domain lies in the larger one, since $\mathrm{dist}(\cdot, S)$ is 1-Lipschitz, so Cauchy's formula costs $1/\sigma$ in each of the D directions (in the angles, $s|k_j|e^{-\sigma s|k|_1} \leq 1/(e\sigma)$), which gives (4.7). □

Lemma 4.4 (Lie step)**.** *Let $|k \cdot \omega| \geq \gamma|k|_1^{-\tau}$ for $k \in \mathbb{Z}^n \setminus \{0\}$, $h_0 = \omega \cdot (\Lambda - \Lambda^0) + c_0$, and $H = h_0 + g + f$ real holomorphic on $\mathcal{D}_\rho$ with $\partial_\lambda g = 0$. Put $\mathcal{E} = \|X_f\|_\rho$, $\mathcal{G} = \|X_g\|_\rho$ and $A_\gamma = \gamma^{-1}\max\{1, (2\tau/(es))^\tau\}$. For $0 < \sigma < \rho - 1$ there is a real χ with*

$$\omega \cdot \partial_\lambda\chi = f - \langle f\rangle, \qquad \|X_\chi\|_{\rho-\sigma/2} \leq A_\gamma\sigma^{-\tau}\mathcal{E} =: u, \tag{4.8}$$

$\langle f\rangle$ the λ-average. There are $c_\mathsf{D}, B_\mathsf{D} > 0$ depending only on D such that, if $u/\sigma \leq c_\mathsf{D}$, the time-one map ϕ of X_χ is real, holomorphic, symplectic and periodic on $\mathcal{D}_{\rho-\sigma}$, moves points by at most u in scaled coordinates, and

$$H \circ \phi = h_0 + g + \langle f\rangle + f_+, \qquad \|X_{f_+}\|_{\rho-\sigma} \leq B_\mathsf{D}\frac{u}{\sigma}(\mathcal{G} + \mathcal{E}). \tag{4.9}$$

The constants do not depend on r, b, S, Λ^0.

Sketch. Put $\chi_k = f_k/(ik \cdot \omega)$ for $k \neq 0$; the divisor is constant, and $|k|_1^\tau e^{-\sigma s|k|_1/2} \leq (2\tau/(e\sigma s))^\tau$ gives (4.8). Iterating (4.7) gives $\|\mathrm{ad}^j_{X_\chi}V\|_{\rho-\eta}/j! \leq (2e\mathsf{D}u/\eta)^j\|V\|_\rho$, so the Lie series converges and represents the flow. The identity $\{h_0, \chi\} = -(f - \langle f\rangle)$ removes X_{h_0}, which is not small, exactly; the remaining terms are at least linear in χ and give (4.9). □

Lemma 4.5 (Normal form on the full slow tube)**.** *In the setting of Lemma 4.4 let $H = h_0 + F$ with $\|X_F\|_3 \le \epsilon$, and let $N \ge 1$ be an integer. There is $q_{\mathsf{D}} > 0$, depending only on D, such that if*

$$A_\gamma \, \epsilon \, N^{\tau+1} \le q_{\mathsf{D}}, \tag{4.10}$$

there is a real holomorphic symplectic map $\Phi : \mathcal{D}_2 \to \mathcal{D}_3$ with

$$H \circ \Phi = h_0 + Y + R, \qquad \partial_\lambda Y = 0, \qquad \|X_R\|_2 \le \epsilon 2^{-N}, \qquad \|X_Y\|_2 \le 2\epsilon. \tag{4.11}$$

The map Φ moves points by at most $1/16$ in scaled coordinates; its inverse is defined on $\mathcal{D}_{3/2}$, with values in $\mathcal{D}_2$ and the same bound; in particular both move actions by at most $r/16$. No condition is imposed on the slow dynamics.

Sketch. Apply Lemma 4.4 N times with $\sigma = 1/N$ on $\rho_j = 3 - j/N$, from $g_0 = 0$, $f_0 = F$. If $\mathcal{E}_j \le \epsilon 2^{-j}$ and $\mathcal{G}_j \le 2\epsilon$, then $u_j \le A_\gamma N^\tau \epsilon 2^{-j}$ and $\mathcal{E}_{j+1} \le 3B_{\mathsf{D}} A_\gamma \epsilon N^{\tau+1} \mathcal{E}_j \le \mathcal{E}_j/2$ for $q_{\mathsf{D}} = \min(c_{\mathsf{D}}, 1/(6B_{\mathsf{D}}), 1/64)$, while $\mathcal{G}_{j+1} \le \sum_{l \le j} \mathcal{E}_l \le 2\epsilon$. The total displacement $\sum_j u_j \le 2A_\gamma \epsilon N^\tau \le 1/32$ is smaller than every allotted loss, so the composition of the time-one maps is defined on $\mathcal{D}_2$ and inverted by the time-reversed flows. □

4.3 Scaling and the exponential time

We apply Lemma 4.5 in the chart of Lemma 4.2 with $n_s = n$, $S = \mathcal{Z}_0(\Lambda^0)$, the fixed widths s, b, and $r = \sqrt{\mu}$. Let L_ω be a Lipschitz bound for $\omega = \nabla h$ near U.

Lemma 4.6 (Scaling)**.** *There are $M_1, \kappa, \mu_1, c > 0$, depending only on $(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$, such that for every $\Lambda^0 \in \mathcal{L}_\delta$ satisfying (2.6) and every $0 < \mu \le \mu_1$ the following holds. Put*

$$r = \sqrt{\mu}, \qquad \vartheta = \frac{1}{2(\tau+1)}, \qquad N = \lfloor \kappa \mu^{-\vartheta} \rfloor, \qquad \epsilon = M_1 \sqrt{\mu}, \tag{4.12}$$

and $h_0 = h(\Lambda^0) + \omega(\Lambda^0) \cdot (\Lambda - \Lambda^0)$. Then $N \ge 1$, $\|X_{H_\mu - h_0}\|_3 \le \epsilon$, and (4.10) holds. In the transformed variables $(\tilde\Lambda, \tilde\lambda, \tilde z)$,

$$H_\mu \circ \Phi = h(\tilde\Lambda) + Y'(\tilde\Lambda, \tilde z) + R, \qquad \Big|\frac{d\tilde\Lambda}{dt}\Big|_\infty \le r\epsilon 2^{-N} \tag{4.13}$$

along every transformed solution in $\mathcal{D}_2$. With $T_\mu = \exp(c\mu^{-\vartheta})$, every portion of a solution of (2.1) lying in $\mathcal{D}_1$, of duration at most T_μ and starting at action Λ^0, satisfies $\|\Lambda(t) - \Lambda^0\|_\infty \le r/4$.

Proof. Shrink s, b so that f and its first derivatives are bounded by M on $\mathcal{D}_4$, and take μ_1 so small that $4r < r_*$ and $r \le 1$. On $\mathcal{D}_4$, the scaled field of $h - h_0$ has only angle components $(\omega(\Lambda) - \omega(\Lambda^0))/s$, of size at most $4L_\omega r/s$, while those of $X_{\mu f}$ are $\mu \partial_\lambda f/r$, $\mu \partial_\Lambda f/s$ and $\mu J \partial_z f/b$, of size at most $M\mu(1/r + 1/s + 1/b)$. The total $4L_\omega r/s + M\mu/r + O(\mu)$ is of order $\sqrt{\mu}$ at the balancing choice $r = \sqrt{\mu}$. Lemma 4.3 converts this into $\|X_{H_\mu - h_0}\|_3 \le M_1\sqrt{\mu}$, losing only the fixed strip between $\mathcal{D}_4$ and $\mathcal{D}_3$.

Condition (4.10) now reads $A_\gamma M_1 \sqrt{\mu}\, N^{\tau+1} \le q_{\mathsf{D}}$, so the admissible order is $N \asymp \mu^{-1/(2(\tau+1))}$. Precisely, choose κ with $A_\gamma M_1 \kappa^{\tau+1} \le q_{\mathsf{D}}/2$; since $\vartheta(\tau+1) = 1/2$, we get $A_\gamma \epsilon N^{\tau+1} \le A_\gamma M_1 \sqrt{\mu}\, \kappa^{\tau+1} \mu^{-1/2} \le q_{\mathsf{D}}/2$. Decrease μ_1 so that $\kappa\mu^{-\vartheta} \ge 2$, hence $N \ge \kappa\mu^{-\vartheta}/2$, and $\epsilon \le \frac{1}{8}$. Lemma 4.5 gives Φ; with $Y' = h_0 - h + Y$ we obtain (4.13), since $d\tilde\Lambda/dt = -\partial_{\tilde\lambda} R$ has scaled size at most $\|X_R\|_2 \le \epsilon 2^{-N}$. Only the divisors $k \cdot \omega(\Lambda^0)$ were inverted; no arithmetic condition is imposed at later times.

Set $c = (\log 2)\kappa/4$, so $T_\mu = 2^{\kappa\mu^{-\vartheta}/4} \le 2^{N/2}$. As Φ is symplectic, Φ^{-1} maps a solution in $\mathcal{D}_1$ to a transformed solution in $\mathcal{D}_2$; adding the displacements of Φ^{-1} at time 0 and of Φ at time t to the drift (4.13), for $|t| \le T_\mu$,

$$\|\Lambda(t) - \Lambda^0\|_\infty \le \frac{r}{16} + r\epsilon|t|2^{-N} + \frac{r}{16} \le \frac{r}{8} + r\epsilon 2^{-N/2} \le \frac{r}{4}. \qquad \square$$

4.4 First exit and the proof of Theorem B

We first compare the canonical elements with the velocity-osculating elements of Section 2; no small divisors enter here.

Lemma 4.7 (Osculating elements). *Let $d \in \{2,3\}$, let $U \subset \mathcal{Q}$ be compact, $m > 0$, and let K be the compact set of states of Lemma 3.5. There are $\mu_2, A_{\rm el}, A_{\rm osc} > 0$, depending only on (n, ν, U, m), such that for $0 < \mu \le \mu_2$ and every state in K we have $E_i^{\rm osc} < 0$, $a_i^{\rm osc} > 0$, $e_i^{\rm osc} < 1$, $r_i^{{\rm osc},-} \le |q_i| \le r_i^{{\rm osc},+}$ and*

$$|a_i^{\rm osc} - a_i| + |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i| \le A_{\rm el}\mu, \qquad |r_i^{{\rm osc},\pm} - r_i^{\pm}| \le A_{\rm osc}\mu. \tag{4.14}$$

If moreover $A_{\rm osc}\mu \le d_\star/2$ and $2A_{\rm osc}\mu \le d_{\rm pair}/2$, with $d_\star, d_{\rm pair}$ from Lemma 3.5, then

$$r_i^{{\rm osc},-} \ge d_\star/2, \qquad r_{i+1}^{{\rm osc},-} - r_i^{{\rm osc},+} \ge d_{\rm pair}/2.$$

Sketch. On K, $|p_i| \le \bar{p}$, $|q_i| \ge d_\star$, $E_i \le -1/(2a_{\max})$ and $e_i \le e_* < 1$, by the proof of Lemma 3.5. Then $|V_i - v_i| \le n\bar{p}\mu$ and $|k_i - 1| \le \mu$, and expanding the definitions gives $|E_i^{\rm osc} - E_i| + |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i| \le A_0\mu$. For small μ, $E_i^{\rm osc} \le -1/(4a_{\max})$ bounds $|a_i^{\rm osc} - a_i|$, and $|e_i^{\rm osc} - e_i| \le |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i|$ (also at circular orbits) gives $e_i^{\rm osc} < 1$ and (4.14). Lemma 3.2 with $k = k_i$ and $v = V_i$ bounds $|q_i|$, and the last claim follows from (3.5) and (4.14). $\square$

The first-exit lemma uses the normal form only through one explicit stopped drift estimate, so that it also serves the spatial and resonant problems. A state is *prograde* if $q_i \ne 0$, $E_i < 0$ and all $H_i > 0$ (Lemma 3.3). Recall that U, r_g, Ω_B are fixed by Lemma 3.6 applied to $\mathcal{L}_\delta$, and that $\Lambda_{\max}$, $a_{\min}$ are given by (3.6) on this U.

Lemma 4.8 (First exit). *Let $d \in \{2,3\}$, $T > 0$ and $0 < r \le r_g$ with*

$$nr + \Omega_B(r) \le \delta/4. \tag{4.15}$$

Consider the maximal solution of (2.1) with admissible initial state, $\Lambda^0 = \Lambda(0) \in \mathcal{L}$ and $C(0) \le B(\Lambda^0) - \delta$, and assume the stopped drift estimate*:*

(SD) if $|t| \le T$ and, for every u between 0 and t, the solution exists at time u, is prograde, and satisfies

$$\|\Lambda(u) - \Lambda^0\|_\infty < r, \qquad C(u) < B(\Lambda(u)) - \delta/2, \tag{4.16}$$

then $\|\Lambda(t) - \Lambda^0\|_\infty \le r/2$.

Then the maximal solution exists on $[-T, T]$, and for $|t| \le T$ it is prograde and

$$\Lambda(t) \in U, \qquad \|\Lambda(t) - \Lambda^0\|_\infty \le \frac{r}{2}, \qquad \|a(t) - a(0)\|_\infty \le \frac{\Lambda_{\max}}{\min_i \nu_i^2}\, r,$$

$$|C(t) - C(0)| \le \frac{nr}{2}, \qquad C(t) \le B(\Lambda(t)) - \frac{3\delta}{4},$$

$$r_i^- \ge d_\star, \qquad r_{i+1}^- - r_i^+ \ge d_{\rm pair}, \qquad |q_i| \ge d_\star, \qquad |q_i - q_j| \ge d_{\rm pair} \quad (i \ne j),$$

where $d_\star = a_{\min}\delta^2/(8\Lambda_{\max}^2)$ and $d_{\rm pair}$ are the constants of Lemma 3.5 for U and $m = \delta/2$.

Proof. Since $C(0) \geq 0$, $\Lambda^0 \in \mathcal{L}_\delta$, and $C(0) \leq B(\Lambda^0) - \delta < B_\star(\Lambda^0)$ makes the initial state prograde (Lemma 3.3). The conditions in (4.16) and prograde are open, so they hold on an open interval $\mathcal{J} \ni 0$, the largest on which the solution exists and satisfies them.

Improvement. For $t \in \mathcal{J} \cap [-T, T]$, (SD) gives $\|\Lambda(t) - \Lambda^0\|_\infty \leq r/2$, and (3.7) with $r/2$ and (4.15) give $C(t) \leq B(\Lambda(t)) - 3\delta/4$.

Exit. Suppose $t_* = \sup \mathcal{J} \leq T$. If the maximal solution ends at t_*, this contradicts Lemma 3.6, as $nr + \Omega_B(r) \leq \delta/2$. Otherwise, on $\mathcal{J}$ the actions lie in U, so $E_i = -\nu_i^2/(2\Lambda_i^2) \leq -\min_{\Lambda \in U, i} \nu_i^2/(2\Lambda_i^2) < 0$. By continuity the state at t_* is elliptic, has $\|\Lambda - \Lambda^0\|_\infty \leq r/2$ and $C \leq B(\Lambda) - 3\delta/4$, and is prograde because $H_i \geq \Lambda_i - C \geq 3\delta/4$ (Lemma 3.3). The open conditions then hold near t_*, a contradiction. Backward time is identical, so $[-T, T] \subset \mathcal{J}$.

Conclusions. The bound $|C(t) - C(0)| \leq n\|\Lambda(t) - \Lambda^0\|_\infty$ is (3.2), and $|a_i(\Lambda) - a_i(\Lambda^0)| = |\Lambda_i - \Lambda_i^0||\Lambda_i + \Lambda_i^0|/\nu_i^2$ gives the axis bound. The separations are Lemma 3.5 with $m = \delta/2$, and (3.6) gives $d_\star$. □

Proof of Theorem B. Orientation and chart. Here $\Lambda^0 \in \mathcal{L} \subset \mathcal{Q}$. By Lemma 3.3, $L \neq 0$; a reflection of the plane, applied to positions and momenta, is symplectic and leaves (2.1) and all constants unchanged, so we may assume $L > 0$, which is conserved. Prograde states then have all $L_i > 0$, the product chart of Lemma 4.1 covers them, and (4.3) gives $C = \frac{1}{2}|z|^2$.

Constants. Take μ_1, c from Lemma 4.6 and $\mu_2, A_{\mathrm{el}}, A_{\mathrm{osc}}$ from Lemma 4.7 for U and $m = \delta/2$. Choose $\mu_0 \leq \min(\mu_1, \mu_2)$ such that $r = \sqrt{\mu}$ satisfies $r \leq r_g$, (4.15) and $\Omega_B(r) \leq \delta/16$ for $0 < \mu \leq \mu_0$, and such that $A_{\mathrm{osc}}\mu_0 \leq d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \leq d_{\mathrm{pair}}/2$. These choices precede μ and the initial state. Fix $0 < \mu \leq \mu_0$, $r = \sqrt{\mu}$ and $T = T_\mu$ with $T_0 = 1$.

The stopped drift estimate. Let $|t| \leq T$ and suppose (4.16) holds, with the state prograde, for all u between 0 and t. Then

$$\tfrac{1}{2}|z(u)|^2 = C(u) < B(\Lambda(u)) - \frac{\delta}{2} \leq B(\Lambda^0) + \Omega_B(r) - \frac{\delta}{2} \leq B(\Lambda^0) - \frac{7\delta}{16},$$

so $z(u) \in \mathcal{Z}_0(\Lambda^0)$, and the orbit between 0 and t lies in the real points of $\mathcal{D}_1$. This uses the whole ball (4.4) and only (4.16). Lemma 4.6 gives $\|\Lambda(t) - \Lambda^0\|_\infty \leq r/4 \leq r/2$, which is (SD).

Conclusions. Lemma 4.8 gives existence on $[-T, T]$, the AMD margin $\delta/2$, the drift bounds with $A_\Lambda = 1$ and $A_a = \Lambda_{\max}/\min_i \nu_i^2$, and the separations; uniqueness holds because (2.1) is analytic. Lemma 4.7 gives the osculating conclusions, and Lemma 3.1 the Newtonian reconstruction. Nothing is assumed about the averaged Hamiltonian Y'. □

5 The spatial problem

In space we need a chart regular at coplanar orbits whose slow variables measure the deficit against a fixed plane; that plane is aligned with L only along the actual solution. There are n fast angles and $2n$ slow pairs.

5.1 A regular spatial chart

Fix a unit vector $\mathbf{N} \in \mathbb{R}^3$ and coordinates in which $\mathbf{N} = (0, 0, 1)$. For one planet with parameter $\nu > 0$ the chart has variables $(\Lambda, \lambda, x, y, u, v)$, with $\lambda \in \mathbb{T}$ and

$$\Lambda > \Gamma + Z, \qquad \Gamma = \tfrac{1}{2}(x^2 + y^2), \qquad Z = \tfrac{1}{2}(u^2 + v^2). \tag{5.1}$$

Let (Q,P) be the planar chart of Lemma 4.1 at (Λ,λ,x,y), embedded in the first two coordinate directions, and put

$$t=\frac{(u,-v,0)}{2\sqrt{\Lambda-\Gamma}},\qquad c_t=\sqrt{1-|t|^2},\qquad R=(1-2|t|^2)\,\mathrm{Id}+2\,tt^{\mathsf T}+2c_t\,[t]_\times,\qquad (q,p)=(RQ,RP),\tag{5.2}$$

where $[t]_\times w=t\times w$; as $|t|^2=Z/(2(\Lambda-\Gamma))<1/2$, the roots are positive and analytic.

Lemma 5.1 (Regular spatial Poincaré chart)**.** *For every $\nu>0$, (5.1)–(5.2) define a real analytic diffeomorphism, with λ read modulo 2π, onto the states $(q,p)\in\mathbb{R}^3\times\mathbb{R}^3$ with negative energy and $(q\times p)\cdot\mathbf{N}>0$. It carries $dp\wedge dq$ to $d\Lambda\wedge d\lambda+dx\wedge dy+du\wedge dv$, it is regular at circular and at coplanar orbits, and*

$$\frac{|p|^2}{2\nu}-\frac{\nu}{|q|}=-\frac{\nu^3}{2\Lambda^2},\qquad |q\times p|=\Lambda-\Gamma,\qquad (q\times p)\cdot\mathbf{N}=\Lambda-\Gamma-Z.\tag{5.3}$$

For n planets, the fixed-plane deficit of the product chart is

$$C_{\mathbf{N}}=\sum_i\Lambda_i-L\cdot\mathbf{N}=\tfrac12\sum_i\big(x_i^2+y_i^2+u_i^2+v_i^2\big).\tag{5.4}$$

Sketch. One checks $R\in SO(3)$ from $[t]_\times^2=tt^{\mathsf T}-|t|^2\mathrm{Id}$, and $R\mathbf{N}$ has third component $1-Z/(\Lambda-\Gamma)$. As $Q\times P=(\Lambda-\Gamma)\mathbf{N}$ (Lemma 4.1), $q\times p=(\Lambda-\Gamma)R\mathbf{N}$; this gives (5.3), and summing over the planets gives (5.4). Where the eccentricity and Z are positive, write $(x,y)=\sqrt{2\Gamma}(\cos\varpi,-\sin\varpi)$ and $(u,v)=\sqrt{2Z}(\cos\Omega,-\sin\Omega)$; then R is a rotation about the line of nodes conjugated by the rotation by Ω about $\mathbf{N}$. A varying rotation adds $(q\times p)\cdot\boldsymbol{\theta}$ to $p\cdot dq$, where $\boldsymbol{\theta}$ is the infinitesimal rotation vector, and the line of nodes is orthogonal to $q\times p$, so $p\cdot dq=\Lambda\,d\lambda+\Gamma\,d(-\varpi)+Z\,d(-\Omega)+d(\cdot)$, with $\Gamma\,d(-\varpi)=\frac12(x\,dy-y\,dx)$ and $Z\,d(-\Omega)=\frac12(u\,dv-v\,du)$. This gives the form on a dense set, and analyticity extends it. For the inverse, $q\times p$ determines u, v, hence R [2]. □

5.2 The spatial domain

Lemma 5.2 (Spatial domain)**.** *Lemma 4.2 holds in space, for the product of the charts of Lemma 5.1, with $2n$ slow pairs $z=(x_i,y_i,u_i,v_i)_i\in\mathbb{R}^{4n}$ and the whole ball*

$$\mathcal{Z}_0(\Lambda^0)=\big\{z\in\mathbb{R}^{4n}:\tfrac12|z|^2\le B(\Lambda^0)-\delta/4\big\};\tag{5.5}$$

its constants do not depend on $\mathbf{N}$. Every state allowed by (2.7) lies in this domain once $\mathbf{N}$ is aligned with its total angular momentum.

Proof. For a real point of (5.5) and a real Λ with $\|\Lambda-\Lambda^0\|_\infty\le 4r_g$, using $\Omega_B(4r_g)\le\delta/16$ and $B\le B_\star$,

$$\Gamma_i+Z_i\le\tfrac12|z|^2\le B(\Lambda^0)-\delta/4\le B(\Lambda)-3\delta/16\le\Lambda_i-3\delta/16.$$

Hence $\Lambda_i-\Gamma_i$, $L_i\cdot\mathbf{N}\ge3\delta/16$ and $|t_i|^2<1/2$ with fixed margins. Moreover $\Lambda_iD(e_i)=\Lambda_i-|L_i|\le\Lambda_i-L_i\cdot\mathbf{N}$, so by (5.4) $\sum_i\Lambda_iD(e_i)\le C_{\mathbf{N}}=\frac12|z|^2\le B(\Lambda)-3\delta/16$. Thus Lemma 3.5 applies, with $m=3\delta/16$, at every point of the ball, also where L is not parallel to $\mathbf{N}$. The rest of the proof of Lemma 4.2 applies verbatim in $6n$ coordinates, and rotation invariance of scalar products and potentials makes the constants independent of $\mathbf{N}$.

For a state allowed by (2.7), $L\neq0$ by Lemma 3.3. Fix $\mathcal{R}\in SO(3)$ with $\mathcal{R}L=|L|\mathbf{N}$; $(q,p)\mapsto(\mathcal{R}q,\mathcal{R}p)$ is symplectic and preserves H_μ and all elements. Along the solution L is conserved, so $C_{\mathbf{N}}=C$, and $L_i\cdot\mathbf{N}>0$ by Lemma 3.3 while $C<B_\star(\Lambda)$. □

5.3 Proofs of Theorem C and Corollary D

Proof of Theorem C. After the rotation of Lemma 5.2, the product chart is symplectic, $H_\mu = h+\mu f$, and $\frac{1}{2}|z|^2 = C$ along the solution. Lemmas 4.3–4.6 hold verbatim with $n_s = 2n$, which changes only constants, not exponents; all small divisors are $k \cdot \omega(\Lambda^0)$, $k \in \mathbb{Z}^n$. The stopped drift hypothesis of Lemma 4.8 is verified as in the proof of Theorem B, with μ_0, $r = \sqrt{\mu}$ and T chosen in the same way from the spatial constants and with the ball (5.5) in place of the planar one. Lemmas 4.8, 3.5 (dimension-free), 4.7 and 3.1 give all conclusions, and undoing the rotation changes no inequality. Corollary D follows, since $\tau = n - 1/2 > n - 1$ and $2(\tau + 1) = 2n + 1$. □

6 Resonant mean motions

The Diophantine condition enters the proofs of Theorems B and C only through the stopped drift estimate of Lemma 4.6; here Nekhoroshev's theorem for a convex Hamiltonian with slow variables replaces it. As explained in Section 1, Theorem E is close to Nekhoroshev's planetary theorem [8, Theorem 12.3]. Table 1 compares Theorem E with Theorems B and C.

6.1 The input theorem

Write points of $\mathbb{R}^{2(n+n_s)}$ as $(\theta, \xi; I, \eta)$, with fast angles θ, fast actions $I \in \mathbb{R}^n$ and slow pairs $(\xi, \eta) \in \mathbb{R}^{n_s} \times \mathbb{R}^{n_s}$, and the form $\sum dI_i \wedge d\theta_i + \sum d\eta_k \wedge d\xi_k$. For a set X and $r > 0$ let $X \ominus r = \{x \in X : \{\|y - x\|_\infty \le r\} \subset X\}$; $\mathfrak{a}_n, \mathfrak{b}_n$ are as in Theorem E.

Proposition 6.1 (Nekhoroshev, definite case with slow variables)**.** *Let $n \ge 2$, $n_s \ge 1$, let $G \subset G' \subset \mathbb{R}^n$ be open and convex, $W \subset \mathbb{R}^{2n_s}$ open and $\rho_* > 0$, with the closed ρ_*-ball about every point of G contained in G'. Let h be holomorphic on $G'_{\rho_*} = \{I \in \mathbb{C}^n : \mathrm{dist}_\infty(I, G') < \rho_*\}$, real on real points, with $\|D^2 h\| \le M_2$ there, and let $g, m_h > 0$ and $\sigma \in \{1, -1\}$ be such that $\|\nabla h(I)\|_\infty \ge g$ and $\sigma D^2 h(I)(v, v) \ge m_h |v|^2$ for all real $I \in G'_{\rho_*}$ and $v \in \mathbb{R}^n$. Then there exists $\varepsilon_* > 0$, depending only on $(n, n_s, G, G', W, \rho_*, h, g, m_h, M_2)$, with the following property.*

Let $0 < \varepsilon < \varepsilon_*$, and let P be holomorphic on the set $\mathcal{F}$ of $(\theta, \xi; I, \eta) \in \mathbb{C}^{2(n+n_s)}$ with $\mathrm{Re}\, I \in G'$, $(\mathrm{Re}\,\xi, \mathrm{Re}\,\eta) \in W$ and all imaginary parts of modulus $< \rho_*$, real on real points, 2π-periodic in each θ_i, with $|P| \le 1$ on $\mathcal{F}$. Let $T_1 > 0$ and let (θ, ξ, I, η) solve Hamilton's equations for $h(I) + \varepsilon P$ on $[0, T_1]$, with $I(t) \in G$ there and $I(0) \in G \ominus \varepsilon^{\mathfrak{b}_n}$. Then $\|I(t) - I(0)\|_\infty \le \frac{1}{2}\varepsilon^{\mathfrak{b}_n}$ for every $t \in [0, \min\{T_1, \varepsilon^{-1} \exp(\varepsilon^{-\mathfrak{a}_n})\}]$ such that $(\xi(u), \eta(u)) \in W \ominus \varepsilon^{\mathfrak{b}_n}$ for all $u \in [0, t]$.

Here ε_* precedes P, T_1 and the solution. This is [8, Theorem 4.4, p. 30], with $\sup|H_1|$ as allowed by [8, Remark 4.5.A], specialized to a definite Hessian, which is quasiconvex with all steepness indices one [8, §1.11, p. 9]; then [8, (4.1), (4.4), (4.5)] give the exponents $\mathfrak{a}_n, \mathfrak{b}_n$. The form is slightly weaker (analyticity on $G' \supset G$, maximum-norm erosions and conclusion, ε_* depending on the domains). It is proved in this form in Lean (Appendix A).

6.2 A finite cover

As ε_* is not quantitative in $G' \times W$, we use finitely many cells, each with a slow ball at its own sharp level. We may assume $\mathcal{L}_\delta \neq \emptyset$, since otherwise no initial state is admissible. Let $n_s = n$ in the plane and $n_s = 2n$ in space, and take U, r_g, Ω_B as in Section 4 and r_*, s, b, M from Lemma 4.2 or 5.2, shrunk so that $|f| \le M$ on the domain (4.5) centred at any $\Lambda^0 \in \mathcal{L}_\delta$. Put $h(\Lambda) = -\sum_i \nu_i^3/(2\Lambda_i^2)$ and

$$w = \min\{r_*, 4s, 4b\}, \qquad r_c = w/4, \qquad \rho_* = w/8, \qquad B_0 = \max\{M, 1\}, \qquad P = f/B_0.$$

Lemma 6.2 (Finite cover)**.** *There are $\bar\Lambda_1,\dots,\bar\Lambda_J\in\mathcal{L}_\delta$ with $\mathcal{L}_\delta\subset\bigcup_j\{\|\Lambda-\bar\Lambda_j\|_\infty<r_c/2\}$ such that the cells $G_j=\{\|\Lambda-\bar\Lambda_j\|_\infty<r_c\}$, $G_j'=\{\|\Lambda-\bar\Lambda_j\|_\infty<2r_c\}$ and $W_j=\{z\in\mathbb{R}^{2n_s}:\frac12|z|^2<B(\bar\Lambda_j)-5\delta/16\}$ satisfy, with constants independent of j:*

(i) $(G_j,G_j',W_j,\rho_*,\pm h)$ satisfy the hypotheses of Proposition 6.1 with $\sigma=\mp1$, and $\pm P$ satisfy them on the domain $\mathcal{F}_j$ built from (G_j',W_j,ρ_*), on whose real points the chart Hamiltonian is $h+B_0\mu P$;

(ii) if $\Lambda^0\in\mathcal{L}_\delta$, $\|\Lambda^0-\bar\Lambda_j\|_\infty<r_c/2$, $\Lambda\in U$ and $\|\Lambda-\Lambda^0\|_\infty\le4r_g$, then $B(\Lambda)\le B(\bar\Lambda_j)+\delta/8$;

(iii) there is $\varepsilon_1>0$ with $\{\frac12|z|^2<B(\bar\Lambda_j)-3\delta/8\}\subset W_j\ominus r$ for all j and $0<r<\varepsilon_1$.

Proof. The cover exists by compactness, and $r_c+\rho_*<2r_c$. (i) Points of $(G_j')_{\rho_*}$ and of $\mathcal{F}_j$ have actions within $2r_c+\rho_*<w\le r_*$ of $\bar\Lambda_j$, so with real parts in U; there h is holomorphic and real, $D^2h=\operatorname{diag}(-3\nu_i^3/\Lambda_i^4)$ is bounded, and at real points $\omega_i\ge g$ and $-D^2h(v,v)\ge m_h|v|^2$ for constants $g,m_h>0$ depending on U and w. Points of $\mathcal{F}_j$ also have angles with imaginary parts below $\rho_*<4s$ and slow coordinates within $\rho_*<4b$ of $W_j\subset\mathcal{Z}_0(\bar\Lambda_j)$, so $\mathcal{F}_j$ lies in the domain (4.5) centred at $\bar\Lambda_j$, and the splitting $H_\mu=h+\mu f$ gives the last claim. (ii) As $r_c/2\le r_g$, $B(\Lambda)\le B(\Lambda^0)+\Omega_B(4r_g)\le B(\bar\Lambda_j)+2\Omega_B(4r_g)\le B(\bar\Lambda_j)+\delta/8$. (iii) With $A_j=B(\bar\Lambda_j)-5\delta/16$ and $A_j'=B(\bar\Lambda_j)-3\delta/8\ge5\delta/8$: if $\frac12|z|^2<A_j'$ and $\|z'-z\|_\infty\le r$, then $|z'|<\sqrt{2A_j'}+\sqrt{2n_s}\,r$; take $\varepsilon_1=\min_j(\sqrt{2A_j}-\sqrt{2A_j'})/\sqrt{2n_s}$. □

6.3 Proof of Theorem E

	Theorems B, C	Theorem E
arithmetic hypothesis	Diophantine (γ,τ) at $\omega(\Lambda(0))$	none
constants depend on	$(n,\nu,\mathcal{A},\delta,\gamma,\tau)$	$(n,\nu,\mathcal{A},\delta)$
time	$T_0\exp(c\mu^{-\vartheta})$, $\vartheta=1/(2(\tau+1))$	$T_0\exp(c\mu^{-\mathfrak{a}_n})$
action, axis, AMD drift	$O(\sqrt{\mu})$	$O(\mu^{\mathfrak{b}_n})$
other conclusions	sharp B, margin $\delta/2$	identical

Table 1: The Diophantine and the resonance-tolerant results; neither implies the other.

Constants. Proposition 6.1 gives, for each j, constants $\varepsilon_j^\pm$ for $\pm h$. Let $r_1>0$ be such that $nr+\Omega_B(r)\le\delta/4$ for $r\le r_1$. Choose $\varepsilon_0\in(0,1]$ with $\varepsilon_0\le\min_j\varepsilon_j^\pm$, $\varepsilon_0^{\mathfrak{b}_n}\le\min\{r_c/4,r_1,r_g\}$ and $\varepsilon_0^{\mathfrak{b}_n}<\varepsilon_1$, and let $\mu_0\le\varepsilon_0/(2B_0)$ satisfy also the conditions of Lemma 4.7. For $0<\mu\le\mu_0$ put $\varepsilon=B_0\mu<\varepsilon_0$, $r=\varepsilon^{\mathfrak{b}_n}$ and $T=\exp(\varepsilon^{-\mathfrak{a}_n})=T_0\exp(c\mu^{-\mathfrak{a}_n})$, with $T_0=1$, $c=B_0^{-\mathfrak{a}_n}$; all this precedes the initial state. Then $r\le r_g$ and $nr+\Omega_B(r)\le\delta/4$.

Stopped drift. We verify the stopped drift hypothesis of Lemma 4.8 for these r and T: if $|t|\le T$ and for all u between 0 and t the state is prograde and satisfies (4.16), then $\|\Lambda(t)-\Lambda^0\|_\infty\le r/2$. Orient or rotate as in Theorems B and C and pass to the chart, where $\frac12|z|^2=C$. As $\Lambda^0\in\mathcal{L}_\delta$, choose j with $\|\Lambda^0-\bar\Lambda_j\|_\infty<r_c/2$. Since $r\le r_c/4$, $\Lambda^0\in G_j\ominus r$ and $\Lambda(u)\in G_j$. By Lemma 6.2(ii),

$$\tfrac12|z(u)|^2=C(u)<B(\Lambda(u))-\delta/2\le B(\bar\Lambda_j)-3\delta/8,$$

so $z(u)\in W_j\ominus r$ by Lemma 6.2(iii). The orbit thus stays in the real points of $\mathcal{F}_j$, where it solves Hamilton's equations for $h+\varepsilon P$. For $t>0$, as $\varepsilon\le1$, $t\le\exp(\varepsilon^{-\mathfrak{a}_n})\le\varepsilon^{-1}\exp(\varepsilon^{-\mathfrak{a}_n})$, and Proposition 6.1 with $T_1=t$ gives $\|\Lambda(t)-\Lambda^0\|_\infty\le\frac12\varepsilon^{\mathfrak{b}_n}=r/2$. For $t<0$, the time-reversed orbit on $[0,-t]$ solves Hamilton's equations for $-h-\varepsilon P$; use ε_j^-.

Conclusion. Lemma 4.8 now gives existence on $[-T,T]$ (uniqueness as in Theorem B) and, on $[-T,T]$, the margin $\delta/2$, the separations, and the drift bounds with $A_\Lambda = B_0^{\mathfrak{b}_n}/2$; Lemmas 4.7 and 3.1 give the rest. In space $n_s = 2n$, and the exponents are unchanged. □

7 Limitations and open questions

The results are asymptotic. The proofs of Theorems B–E do not compute μ_0 or the other constants. They come from compactness: the widths of the complex domain around the whole sublevel, the bound on the complexified perturbation there, the modulus of continuity of B, and, for Theorem E, the thresholds of Proposition 6.1, which are not explicit either. The theorems say nothing about a given value of μ, and no statement about a real planetary system follows from them.

Real masses. An explicit version would need rigorous lower bounds for the complex widths near the boundary of the sublevel, where perihelia and mutual distances approach their minima, rigorous bounds for the complexified perturbation on the whole enlarged slow set, and, for Theorems B and C, control of the small divisors up to a cutoff. With generic majorants one should expect admissible mass ratios many orders of magnitude below planetary values. Tailored estimates do better, but remain far from planetary masses: for a planar Sun–Jupiter–Saturn model near the 5:2 resonance, [3, §5.1, Tables 1 and 3] obtain 5.75×10^{10} years for a perturbation of size $10^{-11.25}$ with initial actions exactly at the resonance, and 1.04×10^{9} years for size $10^{-11.5}$ with nearby initial actions, while the Jupiter–Sun mass ratio is about 10^{-3}. Nor can observations establish a Diophantine condition to all orders, since every interval of mean motions contains resonances of high order; Theorem E removes this hypothesis, but an explicit version would still have to control resonances up to a finite order.

The Solar System. We make no claim about the Solar System. Laskar and Petit classify it as AMD-unstable, following [6], because of the AMD carried by the giant planets [7, §5.4.1], so the hypothesis of our theorems fails for the full eight-planet Newtonian model, whatever the constants. The model (2.1) also omits all non-Newtonian effects.

Open questions.

1. *Explicit constants.* Can the proof be made quantitative, perhaps computer-assisted with a finite Fourier cutoff and rigorous divisor bounds, for a published model of a system with a large AMD margin?
2. *Exponents.* The exponents of Theorem E are those of Nekhoroshev's general theorem, as in [8, Theorem 12.3]. Can they be improved on the sharp sublevel, using the convexity of the Kepler Hamiltonian or the structure of specific resonances?
3. *Other thresholds.* What are the analogues of our theorems for the weaker notion of AMD-stability of [7, §5.3], which allows a collision of the innermost planet with the star?
4. *Longer times.* Does a fixed margin below the strong threshold give stability beyond exponentially long times?

A Formal verification

Theorems A–E, Corollary D and Proposition 6.1 are proved in Lean 4 [5] (version 4.35.0) with the Mathlib library [12], in the repository [1] at commit `74d77b43`. This appendix records what the formal proofs establish, what they do not, and where the statements are.

A.1 What is verified and what is not

The Lean statements are written in the model of Section 2: the exact Hamiltonian (2.1) in heliocentric positions and scaled canonical momenta, the Kepler energies, axes, actions and eccentricity vectors computed from (q, p), the AMD $C = \sum_i \Lambda_i - |L|$, and the thresholds (2.4), defined as infima over the action chamber and proved to be attained. The conclusions include the existence and uniqueness of the solution of Hamilton's equations on $[-T, T]$, and the fact that its barycentric reconstruction solves Newton's equations with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$. All constants are quantified before μ and the initial state. The only additional hypothesis is the membership of $\Lambda(0)$ in the action chamber, which is implied by $a(0) \in \mathcal{A}$ and is needed to state the threshold. The general theorems do not assume $\max_i \nu_i = 1$; the normalized forms are corollaries. Theorems B, C and E are also restated with the AMD and the orbital elements defined only on the states where they make sense (the `_source` versions).

Lean checks that each stated conclusion follows from the stated hypotheses and the definitions, using Mathlib and the axioms listed below. It does not check:

- that the Lean definitions express the intended notions (coordinates, elements, AMD, thresholds, osculating elements); this was checked by comparing every definition and statement with the text, in reviews by readers other than the writers of the formal proofs;
- the identification of B_{ij} with the critical AMD of Laskar and Petit (Lemma 3.4);
- the proofs of Lemmas 4.3–4.6. The formal proofs of Theorems B and C use instead a finite-cutoff normal form of Pöschel type [11] that is proved in the same library, applied on the whole ball (4.4) with fixed widths, a Fourier cutoff of order $\mu^{-\vartheta}$ and an action radius proportional to $\sqrt{\mu}$; it gives the same exponent;
- the citations and the prose.

Proposition 6.1, the input of Theorem E, is itself proved in the same library; its statement was written, from [8, Theorem 4.4], before its proof.

A.2 Statements

Tables 2 and 3 list the formal statements and the principal formal counterparts of lemmas. The declaration names are internal identifiers, listed verbatim, in the namespaces `KAM.Planetary.StrongAMD` (planar), `KAM.Planetary.Spatial` (spatial) and `KAM.Nekhoroshev`; paths are relative to `KAM/Planetary/`, except in the last row of Table 2.

A.3 Axioms and status

`#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for each of the following headline declarations, checked at commit `74d77b43`:

- `lp0_planar_barrier`;

Result	Lean declaration (file and line)
Thm. A spatial	`lp0_planar_barrier` (`StrongAMD/PlanarSafety/Main.lean:255`) `spatialStrongBarrier_physical`, `exists_spatialBarrierCompact` (`Spatial/Geometry.lean:83, 252`)
Thm. B	`lp1_planar_strongAMD_safety`, `_normalized` (`StrongAMD/PlanarSafety/Main.lean:44, 137`); `_source` forms (`Source.lean:38, 132`)
Thm. C	`lp2_spatial_strongAMD_safety`, `_normalized` (`Spatial/SpatialSafety/Main.lean:43, 153`); `_source` forms (`Source.lean:42, 145`)
Cor. D	`lp1_planar_strongAMD_safety_halfInteger` (`StrongAMD/PlanarSafety/Main.lean:190`), `lp2_spatial_strongAMD_safety_halfInteger` (`Spatial/SpatialSafety/Main.lean:210`)
Thm. E	`lp3_planar_strongAMD_safety`, `_normalized` (`StrongAMD/Nek/PlanarSafety/Main.lean:42, 135`; `_source` forms `Source.lean:37, 130`); `lp3_spatial_strongAMD_safety`, `_normalized` (`Spatial/Nek/SpatialSafety/Main.lean:40, 150`; `_source` forms `Source.lean:40, 139`)
Prop. 6.1	`partialConvex_nekhoroshev` (`KAM/Frozen/Nekhoroshev/Theorems/PartialConvexNekhoroshev.lean:48`)

Table 2: Formal statements of the main results, at commit `74d77b43`.

Lemma	Lean declarations (file and line)
3.4	`exists_pairThreshold_attained` (`StrongAMD/Threshold/Pair.lean:97`), `continuousOn_strongThreshold` (`StrongAMD/Threshold/Strong.lean:120`)
3.6	`strongAMD_le_of_tube`, `maximal_global_of_barrier` (`StrongAMD/Continuation.lean:43, 72`)
4.1	`KAM.Planetary.poincare_symplectic` (`Poincare/Symplectic.lean:642`), `KAM.Planetary.poincare_image` (`Poincare/Bijective.lean:320`), `lpChart_strongAMD` (`StrongAMD/PlanarDictionary.lean:156`)
4.2	`lemma8_complete` (`StrongAMD/AnalyticDomain/Bounds.lean:185`)
4.8	`strongFirstExit_of_stopped_drift` (`StrongAMD/Confinement.lean:170`)
4.7	`osculating_barrier` (`StrongAMD/Osculating.lean:268`)
5.1	`spatialPoincare_image` (`Spatial/Chart/Image.lean:180`), `spatialPoincare_symplectic` (`Spatial/Chart/Symplectic.lean:256`), `spatialFixedPlaneDeficit_spatialPoincareProd` (`Spatial/Product/Deficit.lean:110`)
5.2	`lemma18_complete` (`Spatial/AnalyticDomain/Bounds.lean:191`)

Table 3: Formal counterparts of lemmas, at commit `74d77b43`.

- `lp1_planar_strongAMD_safety`, `lp2_spatial_strongAMD_safety` and their `_normalized`, `_halfInteger` and `_source` versions;
- `lp3_planar_strongAMD_safety`, `lp3_spatial_strongAMD_safety` and their `_normalized` and `_source` versions.

No file defining these declarations or imported by them changed through commit `01d36418`. Each integration of new code into the repository also replays the proofs in the Lean kernel. The exact statement texts await the authors' final approval.

A.4 Use of AI and formal tools

The proofs in this paper, their Lean formalization and a first draft of the text were developed with substantial use of large language model systems, working under the direction of the authors. Every formal proof was checked by the Lean kernel. Statements, proofs and the correspondence between the paper and the formal statements were reviewed by independent model-based audits and by the authors. The authors take full responsibility for the content of the paper.

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